

Self-Sorting of White Blood Cells in a Lattice

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When a drop of human blood containing red and white blood cells is forced to move via hydrodynamic forces in a lattice of channels designed to mimic the capillary channels, the white cells self-fractionate into the different types of white cells. The pattern of white cells that forms is due to a combination of stretch-activated adhesion of cells with the walls, stochastic sticking probabilities, and heteroavoidance between granulocytes and lymphocytes. [S0031-9007(97)03796-4]

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A drop of blood contains a complex mixture of red blood cells, which transport oxygen, and white blood cells (leukocytes) which both battle infections and form an essential element of the immune system [1]. Cells respond to both the biological and physical nature of their environments. For the human leukocytes, the environment consists of the human circulatory system and the intercellular spaces which the leukocyte enters from the circulatory system. The adhesion of the leukocyte to the circulatory walls at the capillary stage is absolutely critical for the passage of the leukocyte from the circulatory system into the intercellular space, and occurs in a low Reynolds number hydrodynamic flow. The physiological response of the leukocyte to a shearing flow field is complex: It is known that leukocytes qualitatively change their behavior *after* passing through capillaries [2] and that a complex combination of cell adhesion molecules and threshold hydrodynamic forces are involved in leukocyte adhesion to vessel walls.

We wish here to strip away as much of the biological complexities of leukocyte adhesion as possible and focus on the fundamental aspects of cell adhesion and cell-cell communication. We also wish to avoid much of the fascinating complexity of the hydrodynamics of white cell motion in channels considerably larger than the cell [3] and focus on the cell motions in highly confining lattices smaller in diameter than the unconfined cell [4]. We believe that it is necessary to use hydrodynamic flow to deliver the force [5] rather than static pressure heads using aspiration [6] in order to mimic the physical environment of the cell [7]. By flowing white blood cells through lattices, as shown in Fig. 1, it is possible to subject the cells to a low Reynolds number force field that is similar to the *in vivo* fields and determine their physical response in the absence of biological activators such as *L*-selectin molecules [8,9], and explicitly determine the *physical* aspects of leukocyte adhesion activation. The diameter of a human capillary (4–5 μm) is less than both the aver-

age diameter of a leukocyte *nucleus* (6 μm) and the cell itself (10 μm). We have constructed our array so that the channel width is 5 μm . Our lattice consisted of a simple set of channels of varying length (see Fig. 1). By repeating the channels, we were able to repeatedly strain and release the white blood cells in a physical way related to their passage through the capillary, the massively parallel nature of the lattice enabled us to rapidly get high statistic images of the response of hundreds of white blood cells. Human blood (RHC) was freshly drawn from a finger tip (typical volume was 50 $\mu\ell$), diluted in heparin isotonic buffer to about 10^7 cells/ $\text{m}\ell$, and incubated sequentially with H33342 dye (a vital chromosomal stain [10]) and fluoro-chrome labeled antibodies described below. An inverted microscope (Nikon Microphot., Japan) was used to image

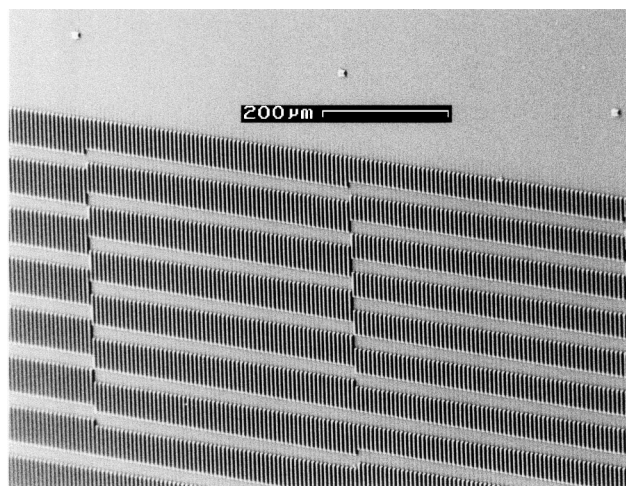


FIG. 1. Scanning electron microscope image of a small section of the variable length array. The total array has channels with lengths varying from 20 to 110 μm in 10 μm steps, with etch depth and width of $5 \times 5 \mu\text{m}$. Fluid flow is from the top down.

the cells, with a Nikon $\times 60$ long-working length objective (MPlan $\times 60$, N.A. 0.7) or a Nikon $\times 4$ objective (Plan-Apo $\times 4$, N.A. 0.20). Chromophore excitation was done either at 350 nm, using a 100 W Hg arc lamp for H33342, or at 488 nm using an Ar⁺ laser (Model 164, Spectra-Physics, Mountain View, CA) for dye antibodies.

While red blood cells readily penetrate and pass through the array the leukocytes are greatly retarded by the channels and eventually stick (adhere) to the surface, although the motion is complex and episodic (stick/slip) in nature as the cells pass through the lattice. Figure 2 shows in epifluorescence the patterns of white blood cells that build up after suction of about 10 $\mu\ell$ of 1/10 (V/V) diluted whole blood at a pressure gradient of approximately 5 (lb/in.²)/in. (or 3×10^3 Pa/cm). There are at least four processes which control cell penetration in the lattice: (1) elastic strain deformation of the cell membrane [6], (2) elastic strain deformation of the nucleus [11,12], (3) surface-specific binding of the cell to the treated silicon dioxide surface, and (4) cell-cell interactions. The first three components do not require a cooperative interaction between the cell types, while the last aspect can be viewed as a simple form of two-body interaction, and results in differential sorting of the cells in the lattice.

The white blood cells form a characteristic pattern by squeezing in an erratic manner through the array and then essentially irreversibly sticking (adhering) at the entrances; they rarely, if ever, adhere in the channels themselves. Cells do not stop within channels where friction would be higher due to increased surface contact with the array surface, but rather stop at the entrance to a channel instead of deforming into it, as is evident from the lines of cells formed. Interestingly, this behavior is independent of channel length L as is clear from Fig. 2. A calculation of the mean penetration length of the cells shows that the mean cell penetration accurately scales with the length. Note that this adhesion occurs *even in the absence* of adhesive protein molecules lining the channel walls. Since the channel surface consists of a polar urethane coating,

we assume that the adhesion is primarily electrostatic in nature, and perhaps related to the diffusion of charged proteins into the contact region [13].

The pressure gradient between the open ends of the channel dominates the driving force on that cell at the entrance to a channel rather than the Stokes drag. Consider the Stokes drag $\mathbf{F}_s$ on a sphere of diameter a moving with speed u in a viscous fluid with viscosity η , equal to $-6\pi\eta a\mathbf{u}$. If a pressure difference ΔP is applied to the length $\mathcal{L}$ of the array, then the pressure gradient Φ is $\Delta P/\mathcal{L}$. If the cell blocks a channel of length L , radius a , then the force F_Φ acting on the cell due to the pressure gradient is equal to $\Phi L\pi a^2$. The ratio $\mathcal{R}$ of the pressure gradient force to the Stokes force is equal to $\Phi La/6\eta u$. In low R_e flow, the average fluid velocity u is proportional to Φ . For our pressure gradients Φ of 1.6×10^5 dyn/cm³, we observe average water velocities u of 100 $\mu\text{m}/\text{sec}$; thus for a 2.5 μm radius channel $\mathcal{R} \sim 10^5 \times L$, independent of Φ . Our shortest channels of 20 μm have $\mathcal{R} \sim 200$. Thus, we can ignore the Stokes drag for cells at entrances to the channels and safely assume that the force acting on the cell is proportional to the channel length. Since the penetration length is proportional to the channel length, it then follows that the sticking force is also proportional to the applied force by a static coefficient of friction κ_s .

Observation of cell nuclear morphology via H33342 dye reveals that different cell types penetrate different distances into the array. We observed that the round, uniformly stained nuclei of the lymphocytes were found deeper in the array than the multilobed nuclei of the granulocytes while the larger diffuse nuclei of the monocytes were found mostly at the array entrance. In order to precisely, and in a parallel manner, ascertain white blood cell identity within the array we incubated whole blood with two cell specific antibodies linked to fluorophores: a fluorescein-labeled antibody CD15 which is specific to most granulocytes (primarily neutrophils) and some monocytes, and a R-phycoerythrin-labeled antibody CD3

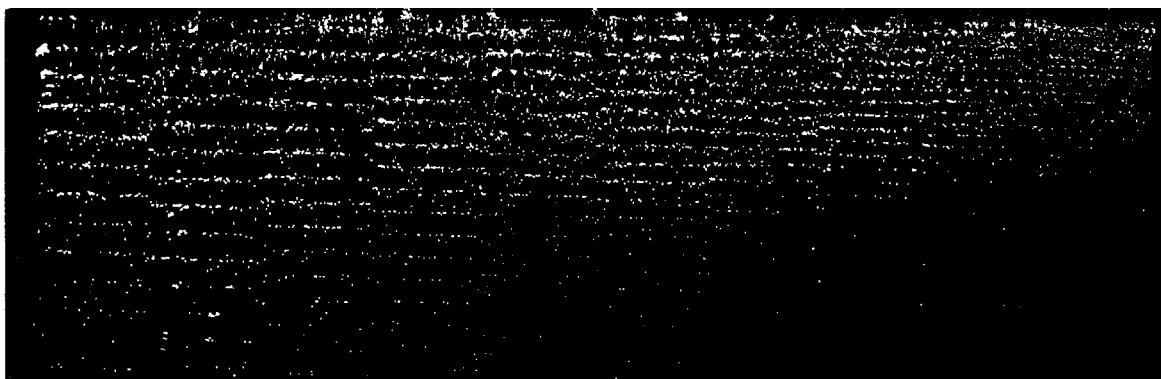


FIG. 2. Epifluorescence picture using the vital nuclear UV dye H33342 of white blood cells in the variable length array, at a lower magnification than Fig. 1 so that all the channels are visible. Fluid flow was from the top down. The epifluorescent images were created after flowing approximately 10 $\mu\ell$ of 1/10 V/V diluted blood through the coated array.

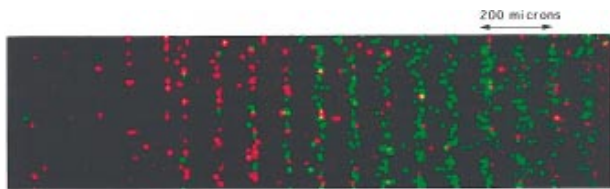


FIG. 3(color). Broad-band antibody color epifluorescence of the captured cells. Flow is from the right to the left. Colors are green fluorescent cell wall (granulocyte and monocyte), orange-red fluorescent cell wall (*T*-lymphocyte), or a dark cell wall (invisible).

which is specific only to *T*-lymphocytes (PharMingen Inc., San Diego, CA 92121).

The two dyes can be excited by the 488 nm line of the Ar^+ laser and are easily distinguished by their corresponding emission colors: fluorescein with emission maximum at 520 nm (green) and *R*-phycoerythrin with emission maximum at 576 nm (orange-red). Therefore each cell with a visible nucleus can be characterized by a green fluorescent cell wall (granulocyte and monocyte), an orange-red fluorescent cell wall (*T*-lymphocyte), or a dark cell wall (invisible). Figure 3 shows the observed distribution of the different cell types as imaged by Kodak RF color film. As predicted from observing nuclear morphologies, *T*-lymphocytes penetrate further than granulocytes and monocytes into the array, and the two cell types with two colors are distributed very differently.

We can model the adhesion of the cells as an *activated* sticking of cells in the array, as would be expected for efficient sticking of leukocytes at the exits of capillaries rather than at the entrances, which would lead to clogging of the capillary. Thus, although we have a synthetic environment, it seems to be displaying aspects of biologically relevant design. Data analysis was done using the software program KALEIDAGRAPH (Synergy Software, Reading, PA). Cell coordinates determined from NIH Image (National Institutes of Health, Bethesda, MD) analysis of the scanned 35 mm film images were binned commensurate with the channels in the image, where we take advantage of the fact that cell penetration is “quantized” in units of channel numbers and not absolute length. Normalized

cell counts with repeated experiments showed binned variances of $\pm 10\%$.

Consider first the granulocytes (green), which form the first wave of cells. In the limit of assuming that the coarse-grained channels can be reduced to a continuum of channels and that stretch activation is proportional to the number of channel passages, we have

$$\frac{dN_g(z)}{dz} = -\alpha z N_g(z) = -\alpha N_g(0) z \exp\left(-\frac{\alpha z^2}{2}\right), \quad (1)$$

where $N_g(z)$ is the number of granulocytes that penetrate to the channel number z (that is, a survival function), $N_g(0)$ is the initial number of cells at $z = 0$, α is the stretch-activated sticking coefficient, and $dN_g(z)/dz$ is the density of stuck cells. Because of the potential confusion between the survival function $N_g(z)$ and the density of stuck cells $dN_g(z)/dz$, we will refer to the density of stuck cells from here on as $\rho_g(z)$. ρ_g is the experimentally observed density of stuck cells per channel number. Figure 4 shows the fit of Eq. (1) to the granulocyte data. The model clearly predicts both the rise and fall in cell density, the initial rise is due to the stretch activation of the sticking coefficient. The most important quantity both physically and biologically is the sticking coefficient α which determines both the rise of the distribution and the fall. We found that for granulocytes in our arrays α is $0.06 \pm 0.01 \text{ channels}^{-2}$.

While the fit of Eq. (1) to the granulocyte density distribution is quite good, we cannot fit Eq. (1) to the *T*-lymphocyte data since it has a more complex initial distribution and too rapid a fall in density to fit Eq. (1). Since the lymphocytes show clear avoidance of the granulocytes, we propose a simple model in which the probability of a lymphocyte sticking to the silicon array surface depends both upon the number of channels penetrated by a lymphocyte (as for granulocytes) and upon the local density of stuck granulocytes, the higher density *inhibiting* the adhesion of lymphocytes, that is, that the cells communicate locally with each other and the effect of signal transduction is to inhibit lymphocyte adhesion. The simplest linear model for lymphocytes adhesion is

$$\rho_T(z) = -\gamma(z)zN_T = -\gamma_0(1 - \beta\rho_g)zN_T, \quad (2)$$

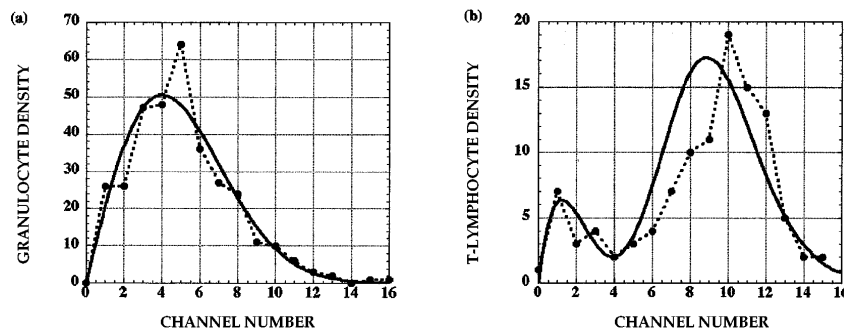


FIG. 4. (a) Fit of Eq. (1) (solid line) to the observed granulocyte density $\rho_g(z)$ (dashed line). (b) Fit of Eq. (3) (solid line) to the observed *T*-lymphocyte density $\rho_T(z)$ (dashed line).

where the T -lymphocyte cell density $\rho_T(z)$ is controlled by a sticking coefficient γ which is locally decreased by the local granulocyte density with the coefficient β . This model is similar to that used to describe the distribution of granulocytes, but now the sticking parameter for lym-

phocytes γ is a function of the distance penetrated into the array and explicitly depends on the local density of stuck granulocytes. Separation of variables and integration by parts yields

$$N_T(z) = N_T(0) \exp \left\{ -\gamma_0 \frac{z^2}{2} + \gamma_0 \alpha \beta N_g(0) \left[-\frac{z}{\alpha} \exp \left(-\frac{\alpha z^2}{2} \right) + \left(\frac{\pi}{2\alpha^3} \right)^{1/2} \operatorname{erf} \left(\left(\frac{\alpha}{2} \right)^{1/2} z \right) \right] \right\}, \quad (3)$$

where $\operatorname{erf}(z)$ is the error function of z . A sixth order polynomial was fit to the error function on the appropriate interval by using the analysis package MATLAB (the Mathworks, Inc., Natick, MA) and used to fit the derivative of Eq. (3) to the observed density ρ_T of stuck T -lymphocytes by using the previously found values for α . The results of that fit are shown in Fig. 4. Note that Eq. (3) correctly predicts the initial "bump" in T -lymphocyte density to adhesion before the granulocyte density has built up, the suppression of the binding over the range of granulocyte adhesion, and the burst of lymphocyte binding after the granulocytes are depleted. The fit value of γ_0 to the T -lymphocytes is 0.05 ± 0.01 channels $^{-2}$, very close to the granulocyte value of 0.06 channels $^{-2}$. Thus we have a prediction that lymphocytes, in the absence of granulocytes ($\beta = 0$), will form a distribution virtually identical to that formed by granulocytes, but, in the presence of granulocytes, the distribution is modulated by cell-cell inhibition of sticking. The repulsive term β has a best fit value of 0.02 ± 0.005 channels/cell. We have carried out an experiment using separated populations of T -lymphocytes and granulocytes and found that, in fact, the cells, in the absence of each other, do form identical distributions fitting Eq. (1).

The success of this simple model indicates that even in the absence of complex biological surfaces we can learn about the physical aspects of cell adhesion to surfaces and the cell-cell interactions which modulate the adhesion of cells at high cell densities. These self-sorting patterns of white blood cells could be used as diagnostic tools of cell disorders, as well as to provide an interesting new laboratory to explore the complex ways that cells respond to the physical aspects of their environment and locally interact with each other. By use of elastomer sealed arrays to which the cells do not stick when the top is removed, we will be able to probe differences among individual stuck cells, either physically, using techniques such as laser tweezers or atomic force microscopy, or via blotting transfer techniques to explore the biochemical variations among cells. It is also possible to mold the structures themselves out of the elastomer or use the elastomer stamping techniques of Whitesides *et al.* [14,15] to create L -selectin or related protein coated surfaces in the array to

probe the biological as well as physical activation of the leukocytes.

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